output/mc_2M_5M/report.txt

```
Baseline, 48-entry write queue (output/mc_2M_5M) - 8 cores
workload        MPKI   WPKI    BLP  write%  WR->WR  WQ occ     WS   | paper:  MPKI  WPKI   BLP write%
fotonik3d_17    25.4    8.1   23.9    30.0   4.93n    22.5   1.76   |         30.6   9.7  23.9   36.9
lbm_17          25.2   19.0   24.2    50.9   4.94n    22.5   1.16   |         48.5  25.5  24.6   51.8
mean            25.3   13.5   24.0    40.5   4.93n    22.5          |  paper avg (all 31 workloads): BLP 22.1, write% 33.0, WR->WR 5.0 ns
n/a = undefined (no write drains, or no single-core run). Means are over workloads with a value: BLP 2 of 2, WR->WR 2 of 2, others 2.
Note: WPKI counts LLC dirty evictions (L3CACHE_WRITEBACKS) only. BARD-C/H cleanses (L3CACHE_BARD_CLEANSES) are also DRAM writes but are not
included in WPKI.

Policies vs baseline, 48-entry write queue (BankRR = round-robin memory controller; speedup = WS / baseline WS - 1; evictions = LRU / BARD-E
override / BARD-C cleanse, % of all evictions)
workload      policy      MPKI   WPKI    BLP  write%  WR->WR  WQ occ   WR lat sameBG%     WS  speedup     LRU%     E%     C%
fotonik3d_17  baseline    25.4    8.1   23.9    30.0   4.93n    22.5   413.2n    0.36   1.76   0.00%    100.0    0.0    0.0
              BARD-H      25.4    0.4   30.3    24.7   4.07n    21.7   379.1n    0.25   1.82   3.18%     67.9    0.4   31.8
              BankRR      25.4    8.1   23.8    30.5   5.11n    22.3   415.5n    0.44   1.75  -0.79%    100.0    0.0    0.0
              RR+BARD-H   25.4    0.4   30.8    25.6   3.95n    21.7   371.8n    0.11   1.83   3.88%     67.8    0.4   31.8
lbm_17        baseline    25.2   19.0   24.2    50.9   4.94n    22.5   257.3n    0.43   1.16   0.00%    100.0    0.0    0.0
              BARD-H      25.2    4.2   31.2    43.7   3.98n    22.5   236.6n    0.20   1.22   5.16%     28.7    6.6   64.6
              BankRR      25.2   19.1   24.2    51.4   5.08n    22.3   259.2n    0.47   1.14  -1.52%    100.0    0.0    0.0
              RR+BARD-H   25.2    4.2   31.2    46.5   4.00n    22.6   233.2n    0.08   1.22   5.08%     28.7    6.7   64.6
Note: WPKI counts dirty evictions only; BARD-C/H cleanses (C%) are DRAM writes too and are not in WPKI.
WR lat = mean time from a write entering the controller's write queue to its WR command (ns); sameBG% = back-to-back WRs in a drain to the
same bank group, % of WR->WR gaps (n/a: older runs).

summary       policy     gmean speedup  mean BLP  mean write%  mean WR->WR  max WR->WR  mean WR lat   (over 2 workloads)
              baseline           0.00%      24.0         40.5        4.93n       4.94n       335.3n
              BARD-H             4.16%      30.7         34.2        4.02n       4.07n       307.8n
              BankRR            -1.16%      24.0         41.0        5.09n       5.11n       337.4n
              RR+BARD-H          4.48%      31.0         36.1        3.98n       4.00n       302.5n
paper (31 workloads; 'BARD' = BARD-H): gmean speedup BARD-E 4.1%, BARD-C 3.3%, BARD-H 4.3% (Fig. 10); BLP 22.1 -> 28.8, write% 33.0 -> 29.3
(Fig. 14);
  WR->WR average 5.0 -> 4.2 ns, Table V 'max' 5.7 -> 5.0 ns; BARD-H evictions 64.7% LRU / 4.8% E / 30.5% C (Fig. 10).
```